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2.1 解：

(a)

$$Z = \overline{\overline{\overline{C+D}+B}+A} \oplus E$$

真值表：

A	B	C	D	E	Z
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	0	1	1	1	0
1	0	0	0	0	1
1	0	0	0	1	0
1	0	0	1	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	0	1	0
1	0	1	1	0	1
1	0	1	1	1	0
0	1	0	0	0	0
0	1	0	0	1	1
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	0	1	1
0	1	1	1	0	0
0	1	1	1	1	1
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	0
1	1	0	1	1	0
1	1	0	1	1	0
1	1	1	0	0	0
1	1	1	0	1	0
1	1	1	1	0	1
1	1	1	1	1	0

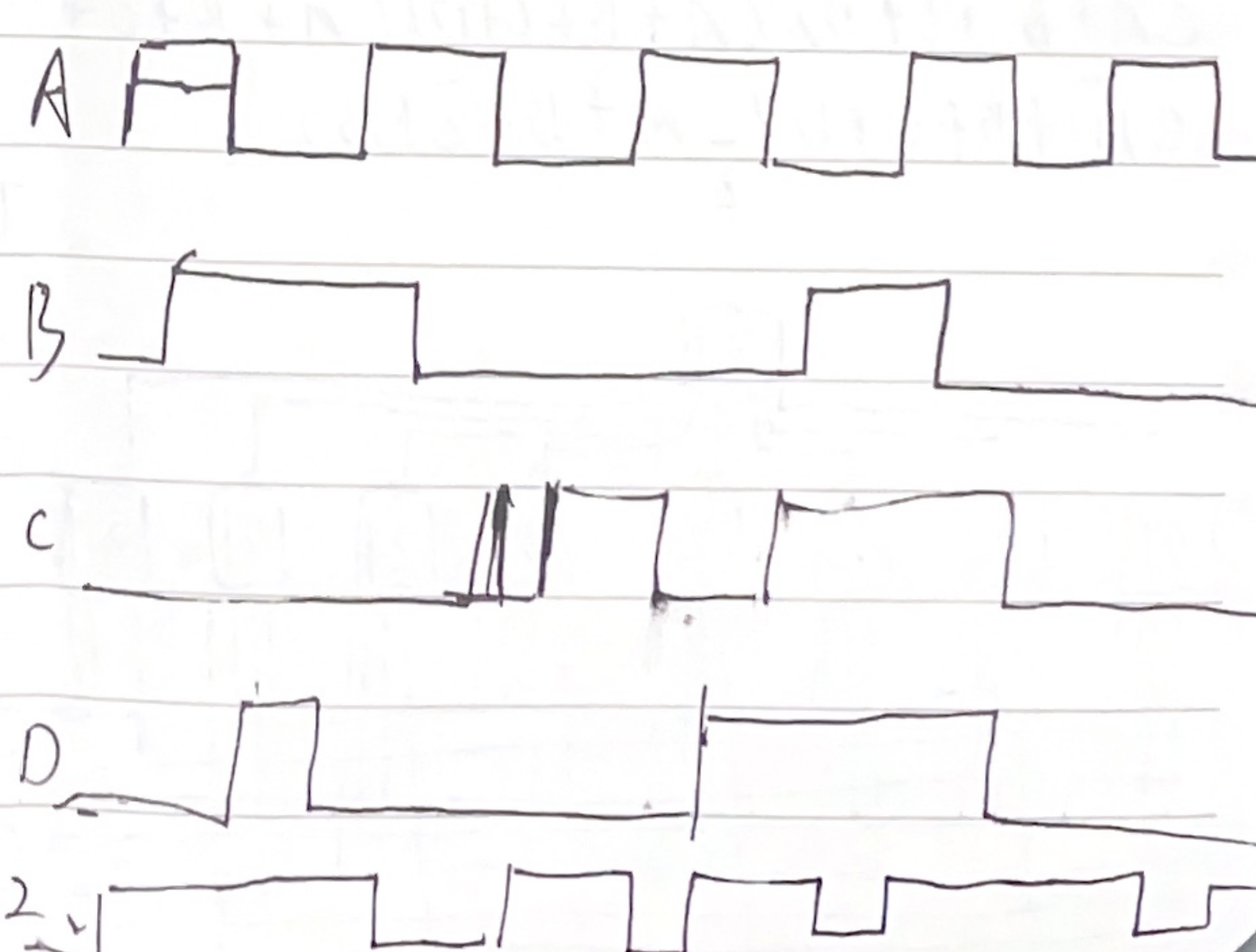
2.3 解：

$$Z = \overline{\overline{\bar{A}\bar{D}B}\ \overline{\bar{B}CD}\ \overline{\bar{A}\bar{D}C}}$$

$$= B\bar{A}\bar{D} + \bar{B}CD + \bar{A}\bar{D}C$$

真值表：

A	B	C	D	Z	A	B	C	D	Z
0	0	0	0	0	1	0	0	0	0
0	0	0	1	1	1	0	0	1	1
0	0	1	0	1	1	0	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	0	1	1	1	0	0	1
0	1	0	1	1	1	1	0	1	1
0	1	1	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	0



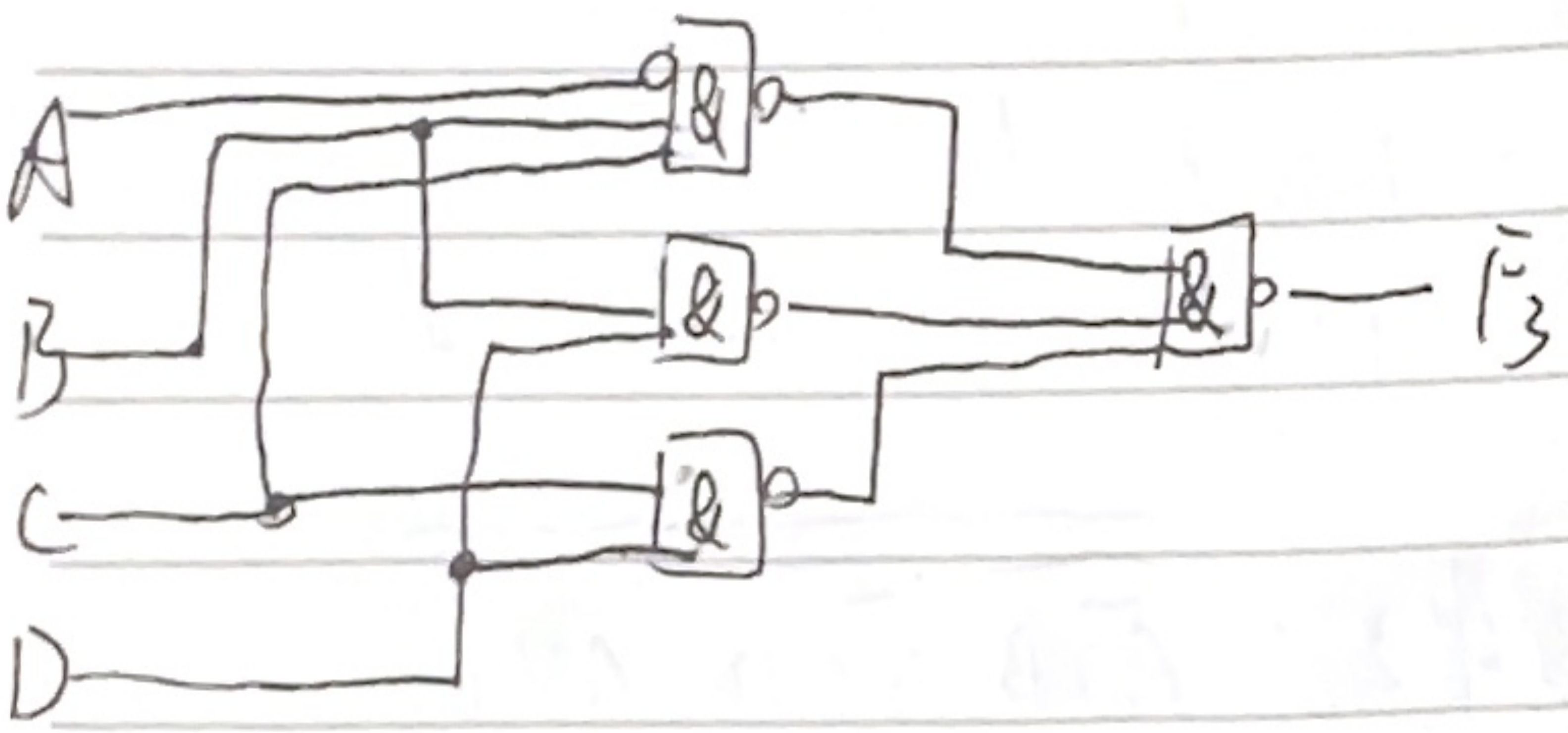
2-5 解：

(3) $F_3 = \bar{A}BC + \bar{A}CD + BD + CD$

$= \bar{A}BC + BD + CD$

$= \overline{\overline{\bar{A}BC + D(B+C)}}$

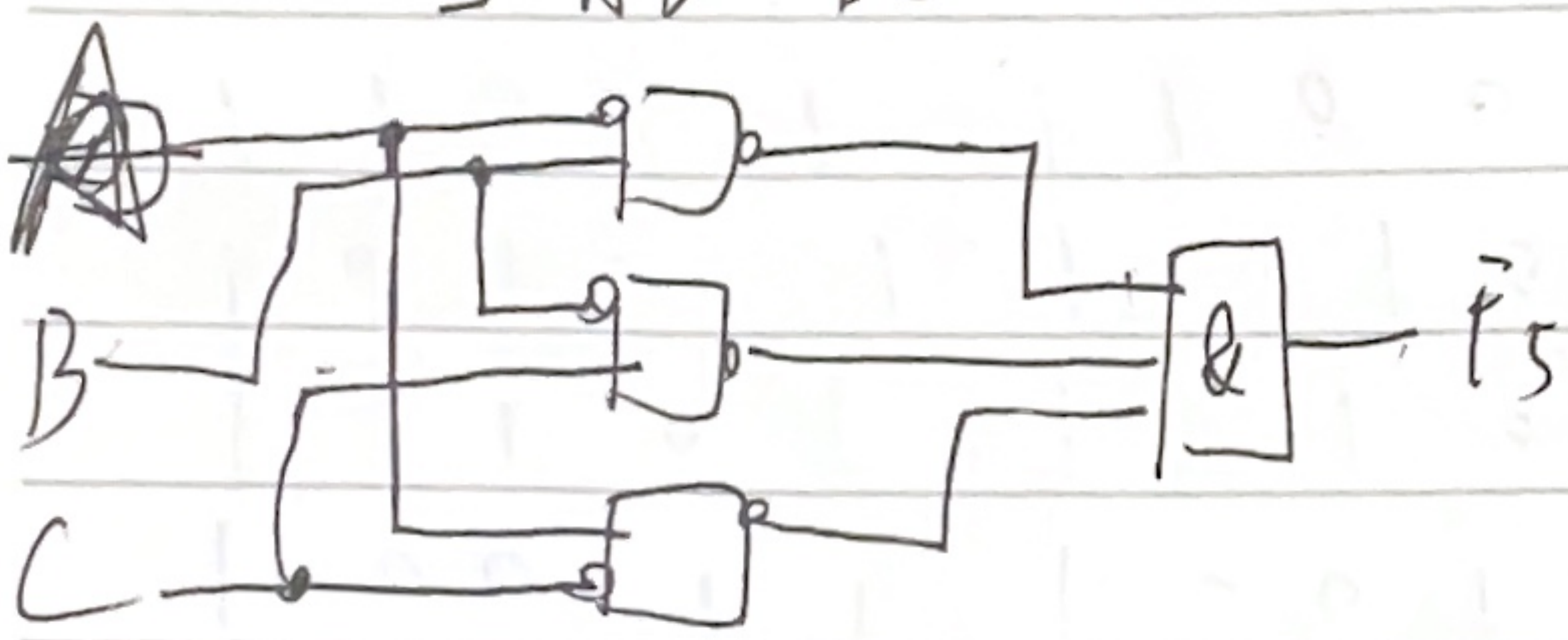
$= \overline{\overline{\bar{A}BC} \cdot \overline{BD} \cdot \overline{CD}}$



(5) $F_5 = (A+\bar{B})(B+\bar{C})(C+\bar{A})$

$= \overline{\bar{A}B + \bar{B}C + \bar{C}A}$

$= \overline{\overline{\bar{A}B} \cdot \overline{\bar{B}C} \cdot \overline{\bar{C}A}}$

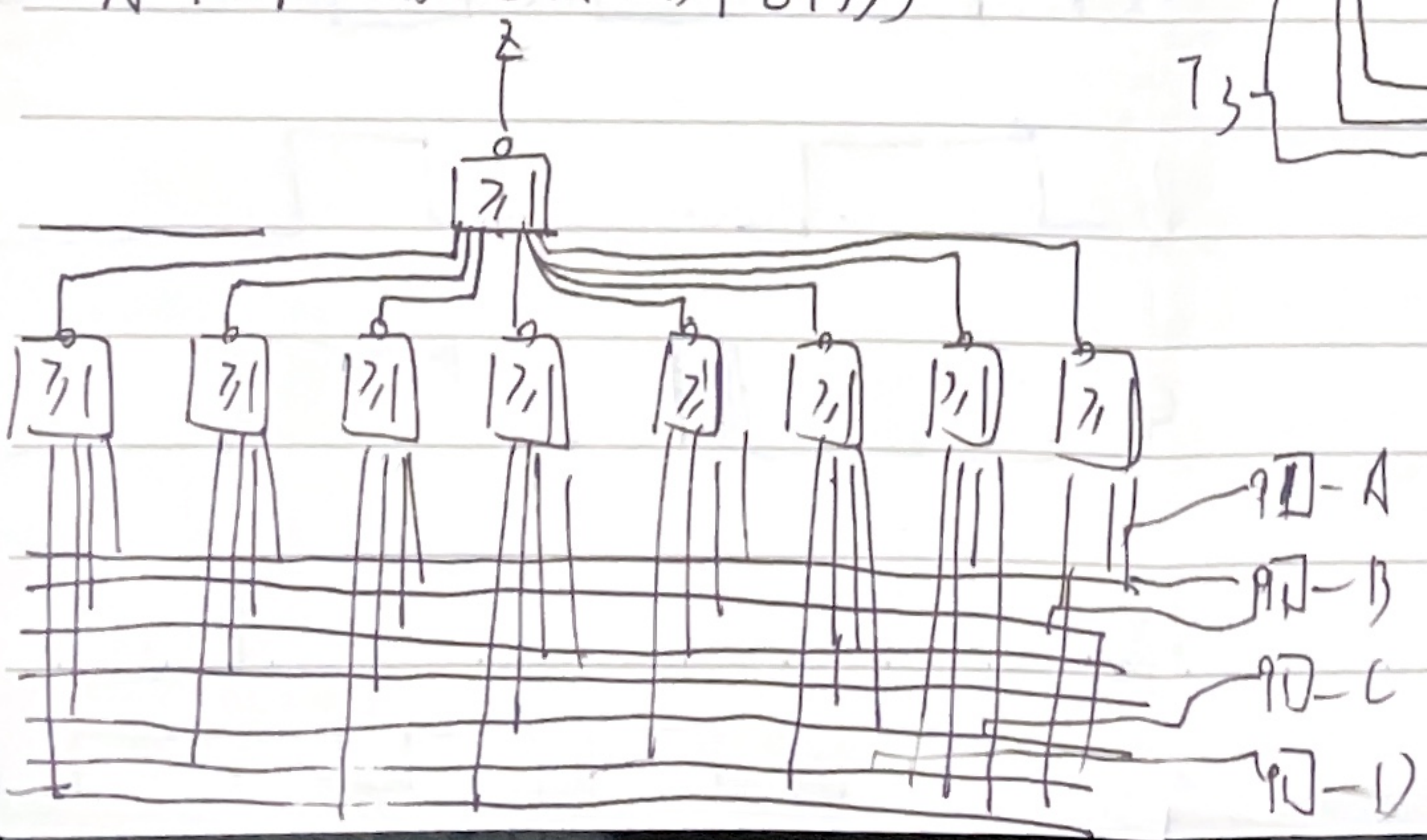


2-7 画出逻图

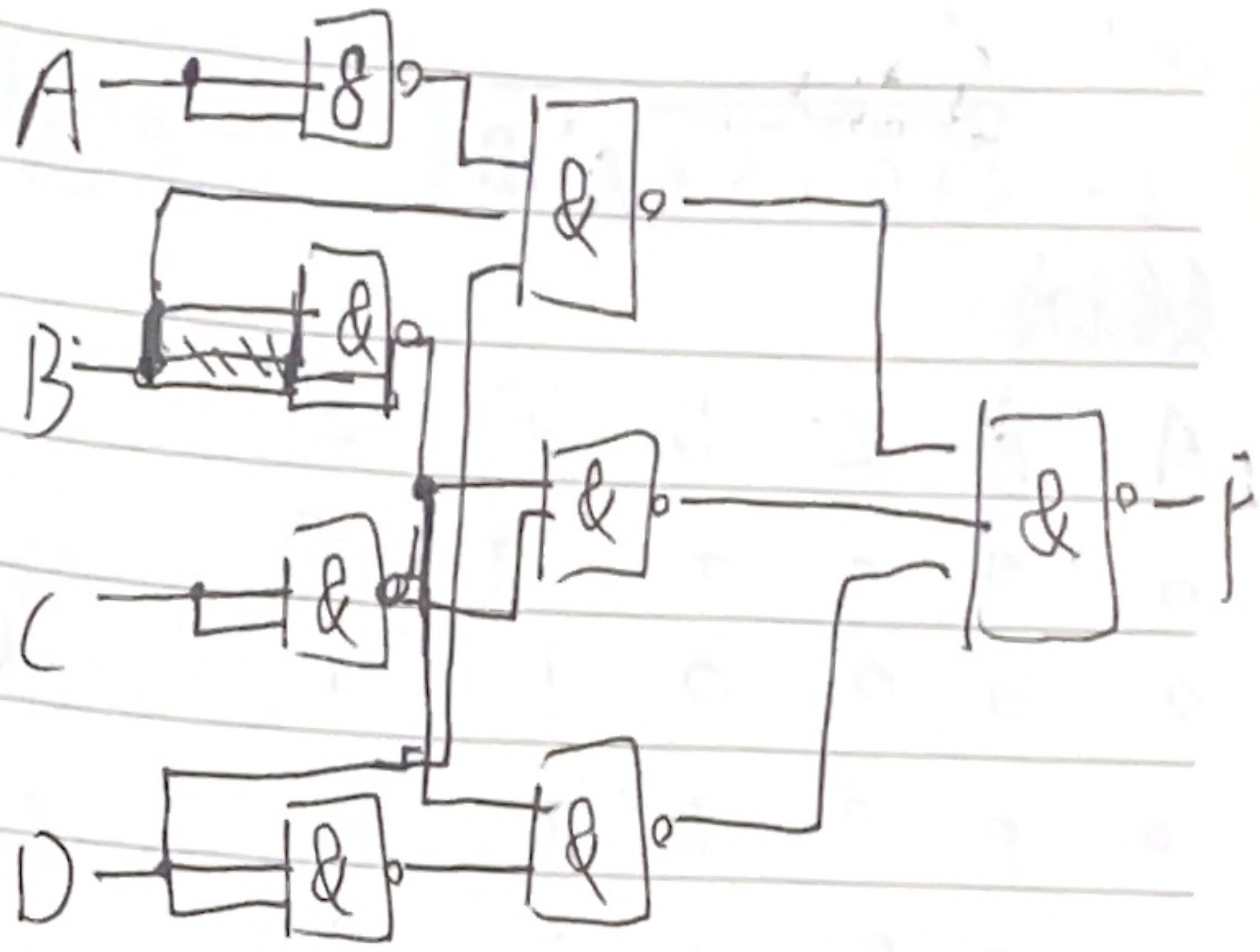
$Z = (A+B+C+D)(A+B+\bar{C}+\bar{D})(A+\bar{B}+C+\bar{D})$

$(A+\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+C+D)(\bar{A}+\bar{B}+\bar{C}+\bar{D})$

$(\bar{A}+B+C+\bar{D})(\bar{A}+B+\bar{C}+D)$



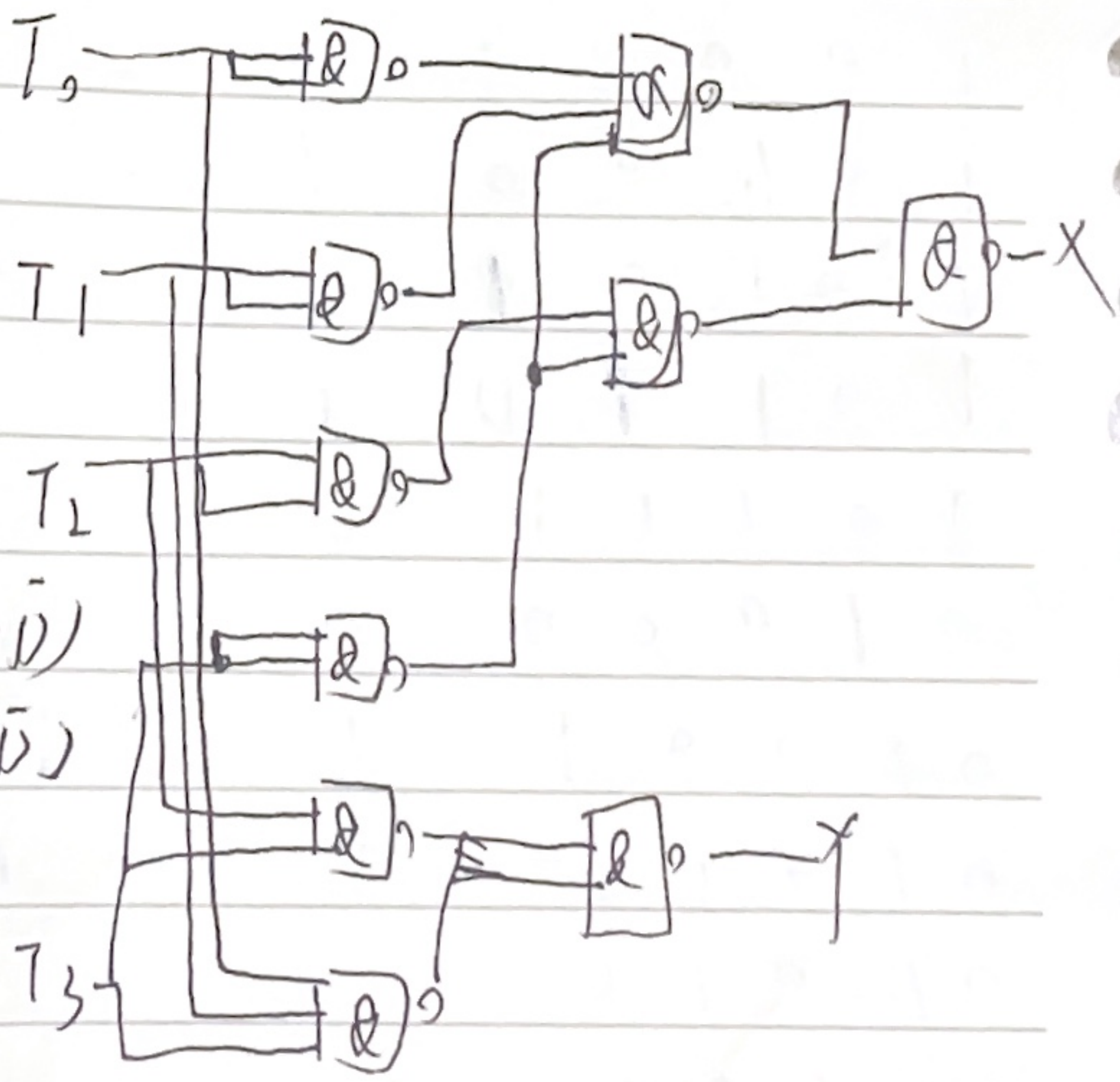
2-9 解：写表达式和逻图



2-13 解：

$X = \overline{\overline{\bar{T}_3\bar{T}_1\bar{T}_0} \cdot \overline{\bar{T}_3\bar{T}_2}}$

$Y = \overline{\overline{T_3T_2} \cdot \overline{T_3T_1T_0}}$



2-15 解：

(a) $Z_1=\overline{\overline{A\bar{B}D}\;\overline{B\bar{C}D}}$

$=A\bar{B}D+B\bar{C}D$

$A=1$，$C=0$，$D=0$ 存在 0 型冒险

(b) $Z_2=\overline{\overline{A+B+\bar{C}}+\overline{B+C+D}+\overline{\bar{A}+\bar{C}+\bar{D}}}$

$=(A+B+\bar{C})(B+C+D)(\bar{A}+\bar{C}+D)$

$B=0$，$C=1$，$D=1$ 或

$A=0$，$B=0$，$D=0$ 时存在 1 型冒险